

# OLED DISPLAYS And Their Applications



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Fig. 1: A flexible OLED display

Over the last few years, organic light-emitting diodes (OLEDs) have found promising applications in flat-panel displays, replacing cathode ray tubes (CRTs) or LED displays. Solid-state OLEDs make it easier to fabricate flexible displays.

The emission of light from materials in the presence of an electric field is called electroluminescence. In 1960s, a single crystal of anthracene helped to observe this phenomenon. Despite the huge quantum efficiency obtained with this organic crystal, no application has been reported with fascinating results. In 1987, Tang and Van Slyke from Kodak achieved efficient and low-voltage OLEDs from a p-n heterostructure device.

OLEDs are thin-film organic semiconductor light-emitting devices. These consist of a thin film of organic material sandwiched between two electrodes (anode and cathode) as shown in Fig. 1. Organic electroluminescent materials are contingent upon pi-conjugated molecules

and are almost insulators. Light is generated when holes and electrons injected at electrodes recombine.

Anode is transparent and made of indium tin oxide. Cathode is reflective and made of metal. When an external potential is applied across electrodes, positive and negative charges (holes from anode and electrons from cathode) are injected. These electrons and holes shift inside the material and recombine to form excitons, emitting photons in the process.

Based on their number of layers, OLEDs can be classified as two-layer and three-layer OLEDs.

In two-layer OLEDs, electrons are injected from cathode into the lowest unoccupied molecular orbital. Simultaneously, holes are injected from anode into the highest occupied molecular orbital.

In three-layer OLEDs, the conductive layer is replaced with electron transport layer and hole transport layer.

Compared to LEDs and LCDs, OLEDs use wide-energy-gap semiconductors and

*An overview of the current status of OLEDs along with different OLED types and their applications*